



USB Device VCP 테스트

[CRZ Technology](#)





Mango-M32F2 USB Device Virtual COM Port 테스트

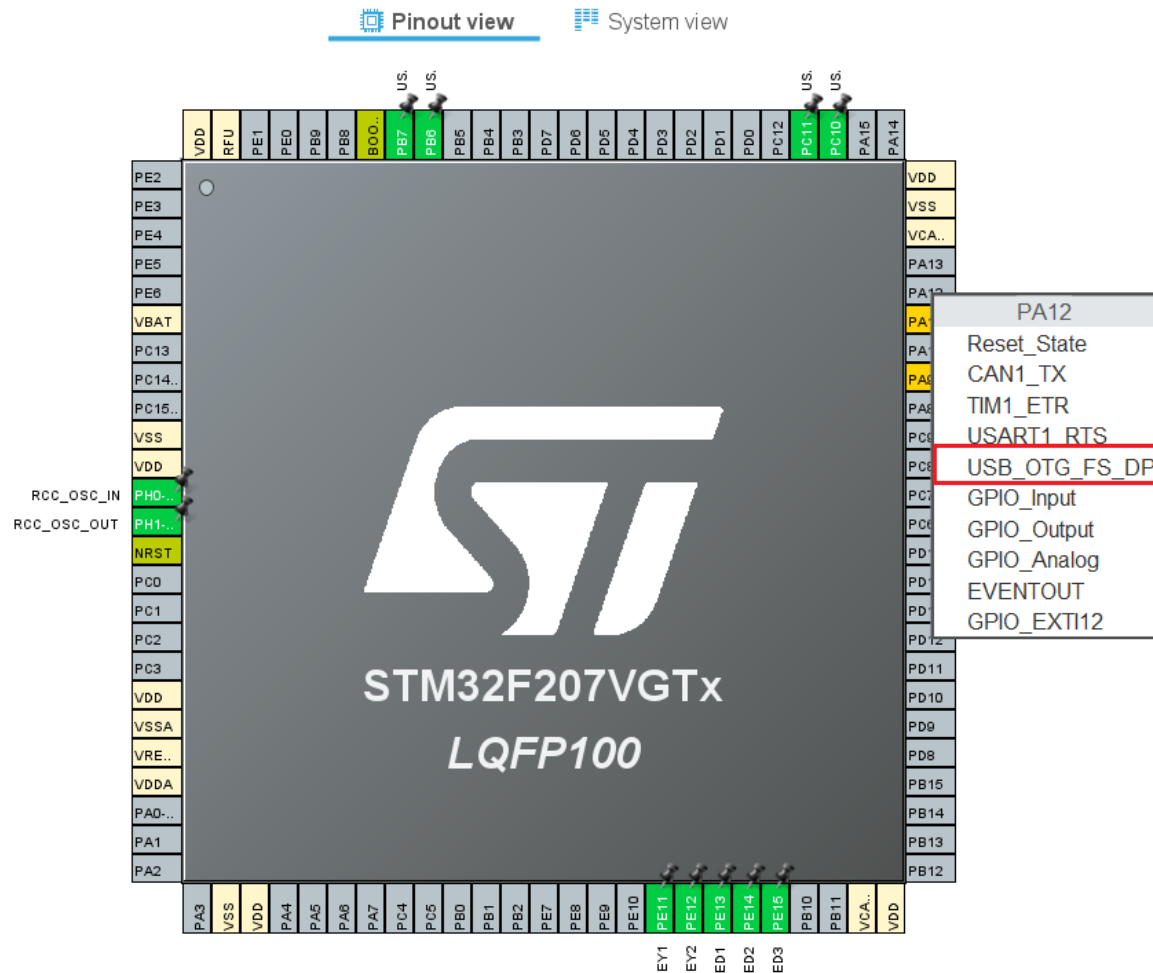
USB Device pinmap

■ USB Device – PA9 / PA11 / PA12

PA8	67	PA8	PA8
PA9	68	PA9	VBUS
PA10	69	PA10	
PA11	70	PA11	OTG_FS_DM
PA12	71	PA12	OTG_FS_DP
PA13	72	PA13	TMS

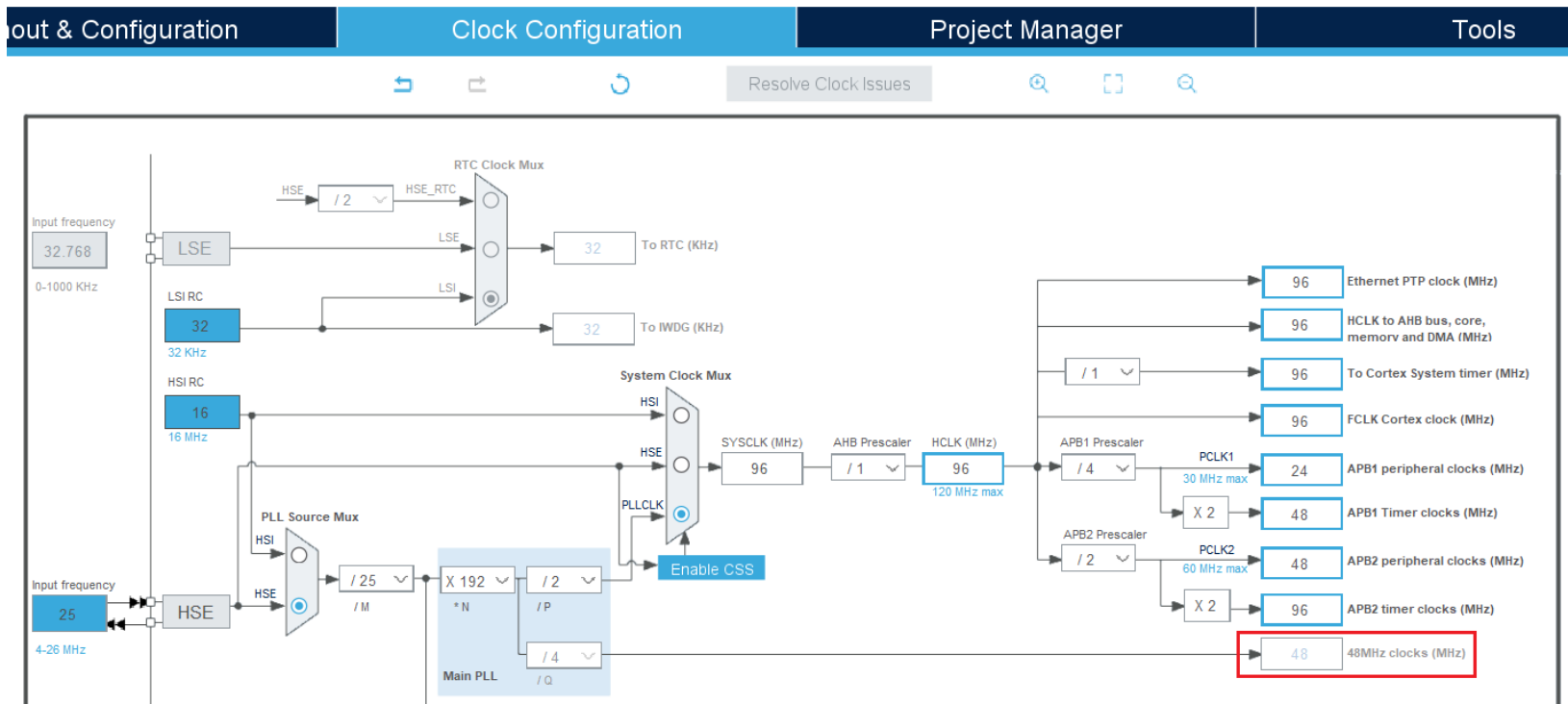
Pinout view

- PA9 – USB_OTG_FS_VBUS, PA11 – USB_OTG_FS_DM
- PA12 – USB_OTG_FS_DP 선택.



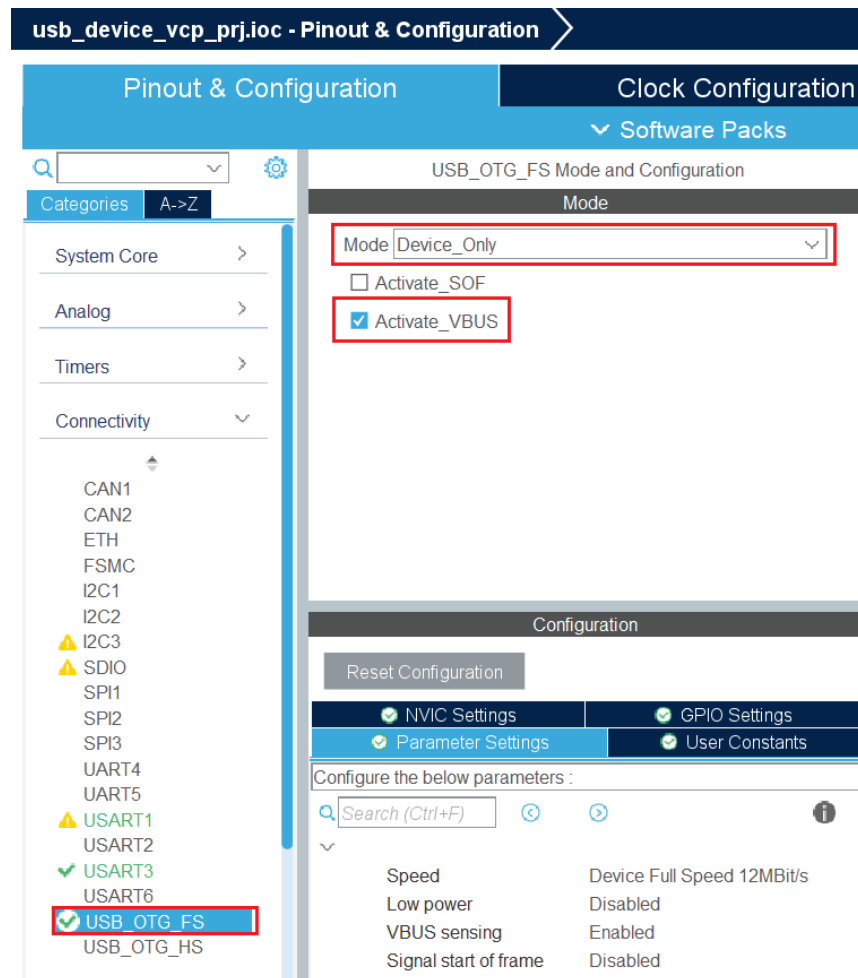
Clock 설정

■ USB 클락 48MHz 생성되도록 설정.



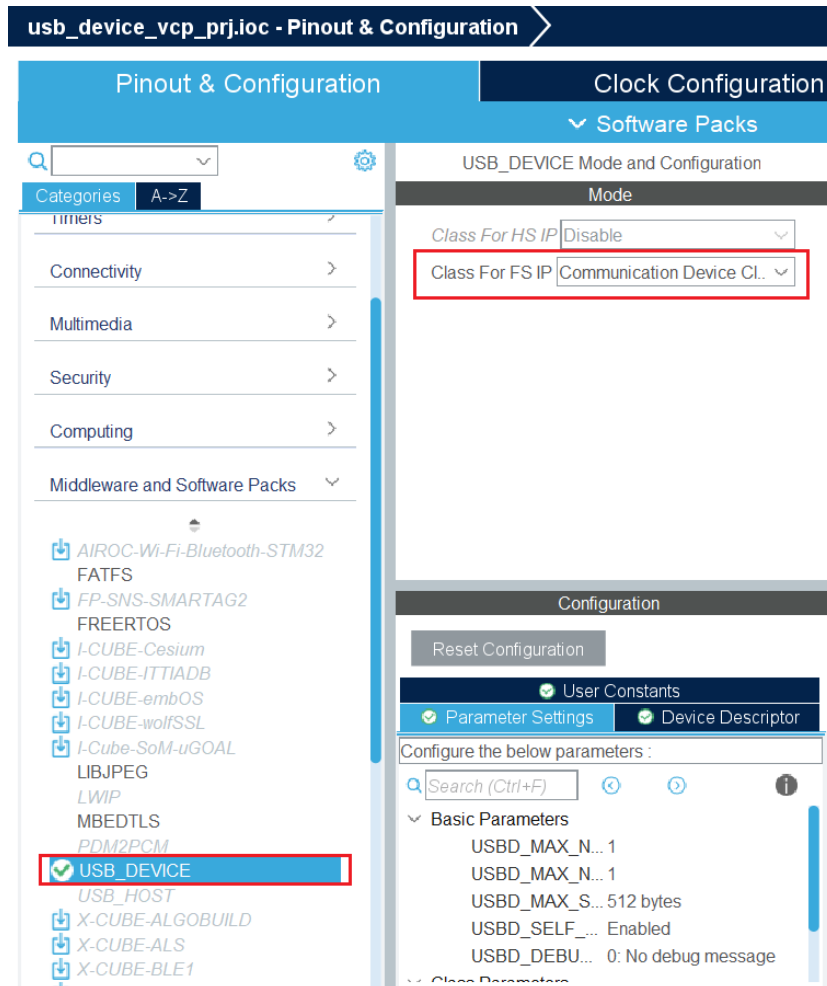
USB_OTG_FS 설정

- “Connectivity” 클릭, “USB_OTG_FS” 클릭.
- “Device_Only” 선택, “Activate_VBUS” 클릭.



USB_DEVICE 설정

- “Middleware and Software Packs” 클릭, “USB_DEVICE” 클릭.
- “Class For FS IP”에서 “Communication Device Class” 선택.



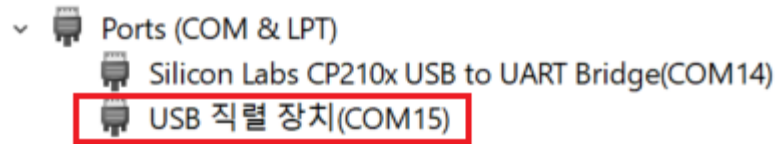
USB Device Virtual COM Port Test Project

- Code 생성 후 Virtual COM Port로 echo 시키기 위해 수정.
- USB_DEVICE/App/usbd_cdc_if.c

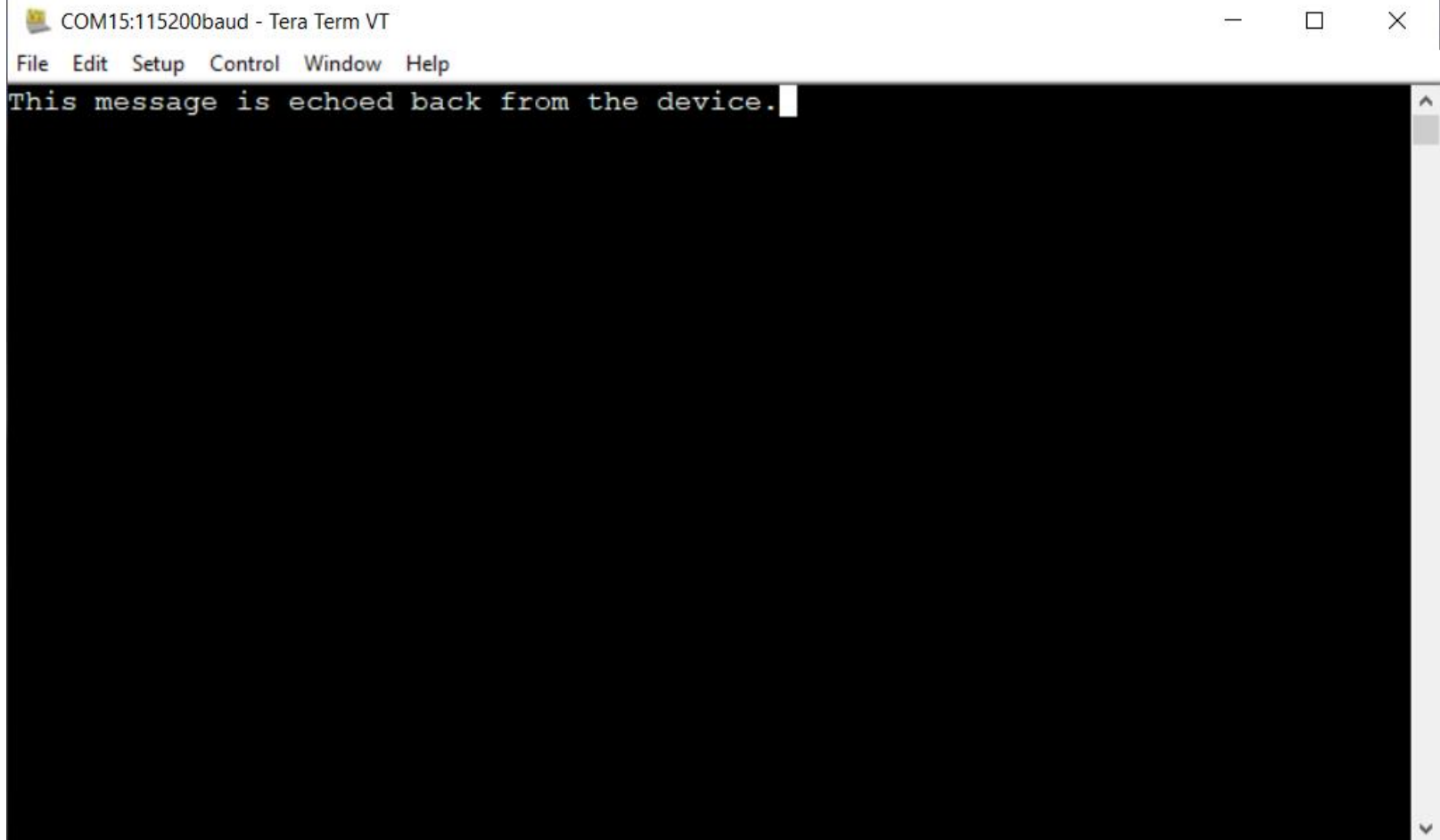
```
259 static int8_t CDC_Receive_FS(uint8_t* Buf, uint32_t *Len)
260 {
261     /* USER CODE BEGIN 6 */
262     USBD_CDC_SetRxBuffer(&hUsbDeviceFS, &Buf[0]);
263     USBD_CDC_ReceivePacket(&hUsbDeviceFS);
264     CDC_Transmit_FS(Buf, *Len);
265     return (USBD_OK);
266     /* USER CODE END 6 */
267 }
```

USB Device Virtual COM Port Test Project

- 이미지 빌드 후 다운로드.
- Mini USB cable을 CON1에 부착 후 PC와 연결 후 부팅.
- COM15로 인식됨.



Test 결과 확인



A screenshot of a Tera Term VT window. The title bar reads "COM15:115200baud - Tera Term VT". The menu bar includes "File", "Edit", "Setup", "Control", "Window", and "Help". The main text area displays "This message is echoed back from the device." followed by a white cursor. The window has standard minimize, maximize, and close buttons in the top right corner.

```
COM15:115200baud - Tera Term VT
File Edit Setup Control Window Help
This message is echoed back from the device. |
```

Thank You

